Lexical exceptionality in paradigm-specific learning:
Modeling stem-final obstruent alternations in Korean verbs and adjectives

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Question

How do learners identify **exceptional** forms and their relevant paradigms when **multiple repair strategies** [1] coexist to solve the same markedness problem?

Korean Stem-Final Conjugation

When /p/-, /t/-, and /s/-final stems conjugate before vowel-initial suffixes, only the irregular (alternating) class undergoes **intervocalic lenition**. Each paradigm adopts a distinct repair strategy:

- p-stems: /p/ → [w] / V_V
- t-stems: /t/ → [ɾ] / V_V
- s-stems: /s/ → ∅ / V_V

*The alternation-triggering environment is absent before consonant-initial suffixes.

Class	Stem	V_V	V_C
p-alt.	/ka.ka**p**/ '*close*'	[ka.ka.**w**-ʌ]	[ka.ka**p**.-ta]
p-non-alt.	/i**p**/ '*wear*'	[i.**p**-ʌ]	[i**p**.-ta]
t-alt.	/kjɛ.da**t**/ '*realize*'	[kjɛ.da.**ɾ**-a.jo]	[kjɛ.da**t**.-ta]
t-non-alt.	/mu**t**/ '*bury*'	[mu.**t**-ʌ]	[mu**t**.-ta]
s-alt.	/na**s**/ '*better*'	[na.-a.do]	[na**s**.-ta]
s-non-alt.	/u**s**/ '*laugh*'	[u.**s**-ʌ]	[u**s**.-ta]

The Indexation Approach

Alternating and non-alternating classes require competing generalizations:

- alt. stems: *V-Obs-V ≫ Faith
- non-alt. stems: Faith ≫ *V-Obs-V

[2] resolves this ranking conflict through constraint cloning [3]:

*V-Obs-V$_{\text{Alter}}$ ≫ Faith ≫ *V-Obs-V$_{\text{Non-Alter}}$

Expanded ranking: Paradigm-specific repairs require different Faith constraints to be outranked by those associated with other alternation patterns.

*V-Obs-V$_P^{Alt}$ ≫ Id-lat, Max ≫ Id-C ≫ *V-Obs-V$_P^{NonAlt}$
*V-Obs-V$_T^{Alt}$ ≫ Id-C, Max ≫ Id-lat ≫ *V-Obs-V$_T^{NonAlt}$
*V-Obs-V$_S^{Alt}$ ≫ Id-C, Id-lat ≫ Max ≫ *V-Obs-V$_S^{NonAlt}$

Scaled MaxEnt

A probabilistic implementation of constraint indexation [4, 5].

$$\mathcal{H}_i = \sum_{\gamma\in\Gamma}\left(w_\gamma + \sum_{m\in\mu_i} s_{\gamma m}\right) v_{\gamma i}$$

(Γ: constraint set; $\mathcal{H}_i$: candidate harmony; w_γ: general constraint weight; $s_{\gamma m}$: morpheme-specific scaling factor; $v_{\gamma i}$: violations of γ by candidate i)

The additive penalty $s_{\gamma m}$ plays a role analogous to lexical indexation (*Predictions*):

(1) *V-Obs-V$_{\text{Alter}}$ ≫ Faith ⇒ increased scaling for alternating classes

(2) Faith constraints implicated in the repair receive the smallest scaling factors:
- Id-C (/p/-paradigm)
- Id-lat (/t/-paradigm)
- Max (/s/-paradigm)

(3) Only lexically idiosyncratic stems receive nonzero scaling; suffixes remain unscaled.

Idealized learning outcome

	*V-Obs-V	Max	Id-lat	Id-C
General weights	4	6	6	6
p-alternating	6	3	2	0
p-non-alternating	0	1	1	1
t-alternating	6	3	0	4
t-non-alternating	0	1	1	1
s-alternating	6	0	3	3
s-non-alternating	0	1	1	1
suffixes	0	0	0	0

Experiment

Toy stems were generated in CV-/p,t,s/ forms, reflecting Korean syllable phonotactics. Frequencies were modeled after the distribution reported in [2].

Suffix inventories were constructed by progressively adding paired -V and -CV suffixes, each either triggering or failing to trigger lenition.

Stem statistics

p-alt	68
p-non-alt	5
t-alt	5
t-non-alt	7
s-alt	5
s-non-alt	4

Toy suffix inventories

Condition	Suffix inventory
2-suffix	*-ta, -a*
4-suffix	*-ta, -a, -te, -e*
6-suffix	*-ta, -a, -te, -e, -tu, -u*
8-suffix	*-ta, -a, -te, -e, -tu, -u, -to, -o*
10-suffix	*-ta, -a, -te, -e, -tu, -u, -to, -o, -ti, -i*

Results

Scaling weights increasingly converge on the locus of exceptionality as the number of suffix types increases:

Stage 1: Scaling weights are misattributed to non-idiosyncratic suffixes.

2-suffix condition

	*V-Obs-V	Max	Id-lat	Id-C
General weights	0.69	2.6	2.6	0
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	0
t-alternating stems	0	0	0	0
t-non-alternating stems	0	0	0	0
s-alternating stems	0	0	0	0
s-non-alternating stems	0	0	0	0
-V suffix	0.69	0	0	0
-CV suffix	0	1.92	1.92	4.51

Stage 2: Scaling weights begin to distinguish alternating and non-alternating stems.

Note: /p/-alternating stems remain unscaled (i.e., fail to be lexicalized).

6-suffix condition

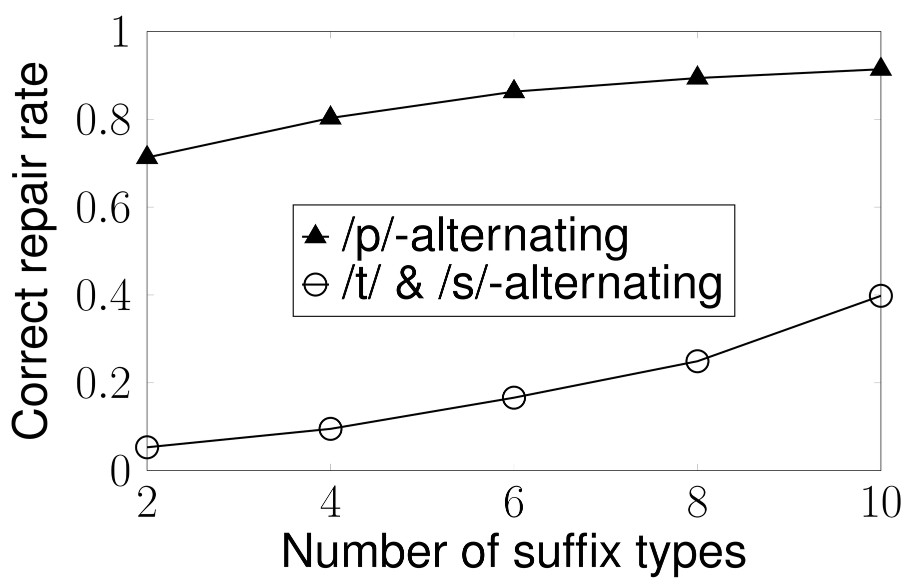
	*V-Obs-V	Max	Id-lat	Id-C
General weights	4.98	6.29	6.29	2.71
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	2.54
t-alternating stems	0.61	0	0	2.88
t-non-alternating stems	0	0	0	2.54
s-alternating stems	0.61	0	0	2.88
s-non-alternating stems	0	0	0	2.54
-V suffix	0	0	0	0
-CV suffix	0	0	0	1.53

Stage 3: Paradigm-specific patterns are distinguished among alternating stems.

10-suffix condition

	*V-Obs-V	Max	Id-lat	Id-C
General weights	7.11	8.98	8.98	4.46
p-alternating stems	0	0	0	0
p-non-alternating stems	0	0	0	3.76
t-alternating stems	2.55	0.69	0	5.2
t-non-alternating stems	0	0	0	3.76
s-alternating stems	2.55	0	0.69	5.2
s-non-alternating stems	0	0	0	3.76
-V suffix	0	0	0	0
-CV suffix	0	0	0	0

This progression coincides with improved prediction accuracy.



Repair accuracy in intervocalic contexts

- Predictions (1) and (2) emerge gradually in the 6-suffix and 8-suffix conditions, respectively.
- Prediction (3) is satisfied only once sufficient morphological evidence becomes available (10-suffix condition).
- Across all conditions, the /p/ → [w] alternation is absorbed into the general grammar rather than being treated as a lexically idiosyncratic pattern.
  - the /p/-alternating class remains unscaled;
  - Id-C receives the largest scaling weights in the remaining classes.
- Distributional asymmetry between /p/ and /t,s/ alternations is reflected in prediction accuracy, with the former learned more reliably.

Discussion

- The model adopts a *least-cost* solution by treating statistically dominant morpheme classes as the general pattern.
- Sensitivity to lexical statistics: without sufficient morphological evidence, the locus of exceptionality cannot be reliably learned.
- Our empirical simulation suggests that, in naturalistic settings, statistical sensitivity arises from limited paradigm saturation [6] rather than absolute type frequency.

References

[1] Joe Pater. Austronesian nasal substitution and other nc effects. The prosody-morphology interface, pages 310–343, 1999.
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[5] Coral Hughto, Andrew Lamont, Brandon Prickett, and Gaja Jarosz. Learning exceptionality and variation with lexically scaled maxent. In Proceedings of the Society for Computation in Linguistics (SCIL) 2019, pages 91–101, 2019.
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Code:
`https://github.com/stxllastar/scaled_conjugation`
Paper: